Reg. No. [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [18EEDE543]

SEVENTH SEMESTER B.Tech EXAMINATION, DECEMBER-2021

SENSORS AND TRANSDUCERS

Time: 3 Hours **Maximum Marks: 70**

Instructions:
i. Answer **FIVE** full Questions. Question number 1 and 2 are compulsory. Each question carries 14 marks.
ii. Missing data may be suitably assumed.

1.	a.	Explain about types and properties of gauge materials.	07
	b.	Explain about Micro Electromechanical Systems (MEMS) its Advantages and applications.	07
2.	a.	Explain classification of transducers.	07
	b.	Explain about frequency domain transducers and digital encoder.	07
3.	a.	Explain the construction of LVDT transducer.	07
	b.	Explain about induction potentiometers and fiber optic transducers.	07
		OR	
4.	a.	Explain about resistance transducers, variable inductance transducers.	07
	b.	Explain about Synchros and Resolvers.	07
5.	a.	Explain about mutual inductance change type sensors.	07
	b.	Explain about variable distance-parallel plate type capacitive sensors.	07
		OR	
6.	a.	Explain about piezoelectric effect & ultrasonic sensors.	07
	b.	Explain about parallel plate, serrated plate/teeth type and cylindrical type sensors.	07
7.	a.	Explain about objectives and configuration of data acquisition system.	07
	b.	Explain about functions of signal conditioning equipment.	07
		OR	
8.	a.	Explain about data conversion.	07
	b.	Explain about data acquisition systems.	07